

Wazuh SIEM Monitoring & Alert Analysis Report

Environment: Kali Linux

Focus: Host-based monitoring, authentication analysis, and threat hunting

Prepared for: Cybersecurity Portfolio & Professional Review

This report documents the deployment and use of the Wazuh Security Information and Event Management (SIEM) platform to monitor, analyze, and interpret security-relevant events on a Kali Linux system. The analysis reflects typical Security Operations Center (SOC) monitoring and alert triage activities.

1. Assessment Objective

The objective of this project is to demonstrate practical SIEM monitoring skills using Wazuh. Specifically, the project aims to collect and analyze system logs, monitor authentication activity, detect privilege escalation events, and evaluate alert severity to support effective SOC decision-making.

2. Tools & Methodology

SIEM Platform: Wazuh

Operating System: Kali Linux

Data Sources: PAM authentication logs, syslog events, sudo activity

The Wazuh agent collected host-based logs from the Kali Linux system and forwarded them to the Wazuh manager. Alerts were generated based on predefined rules and mapped to MITRE ATT&CK; techniques where applicable. The Wazuh dashboard was used for alert review, severity analysis, and threat-hunting visualization.

3. Analysis & Observations

Analysis focused on authentication events, session lifecycle activities, and privilege escalation behavior. The dashboard views provided visibility into alert distribution by severity and technique. Observed alerts primarily represented normal system usage and routine administrative actions, highlighting the importance of contextual analysis in SOC operations.

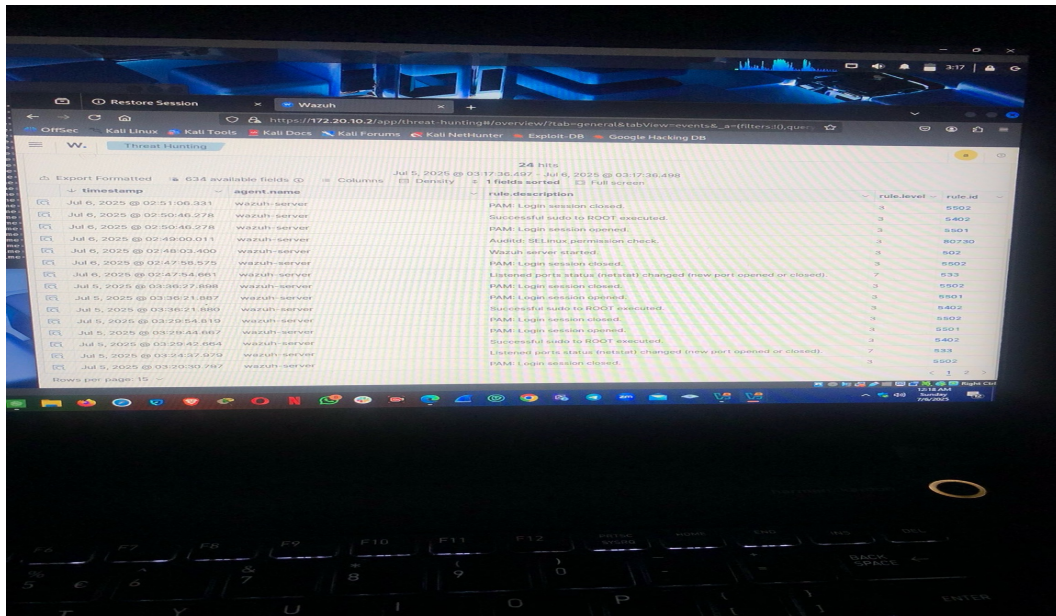


Figure 1: Wazuh threat-hunting dashboard showing authentication-related events.

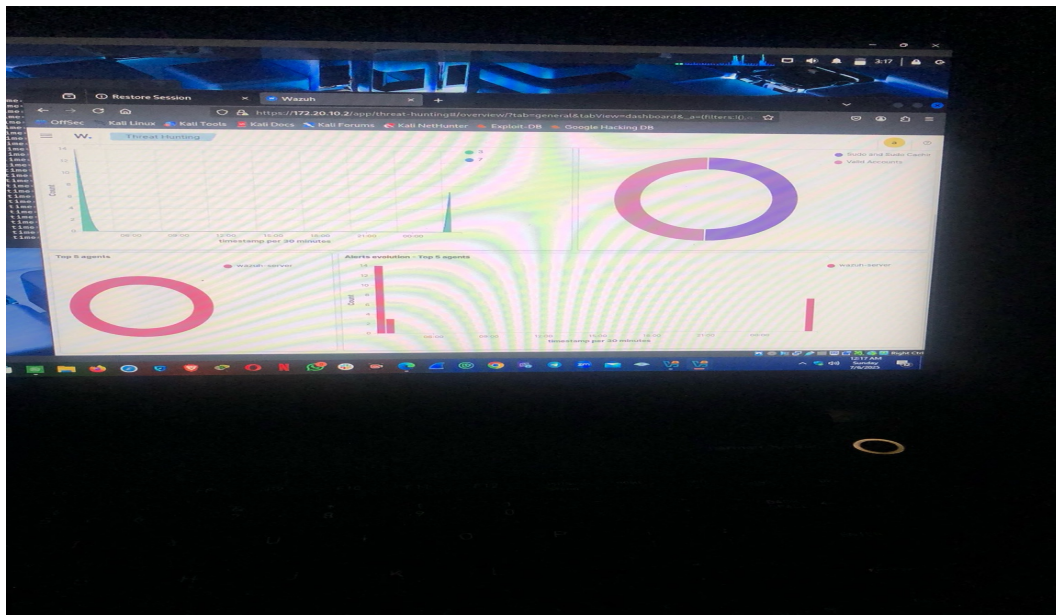


Figure 2: Alert severity distribution and event timeline.

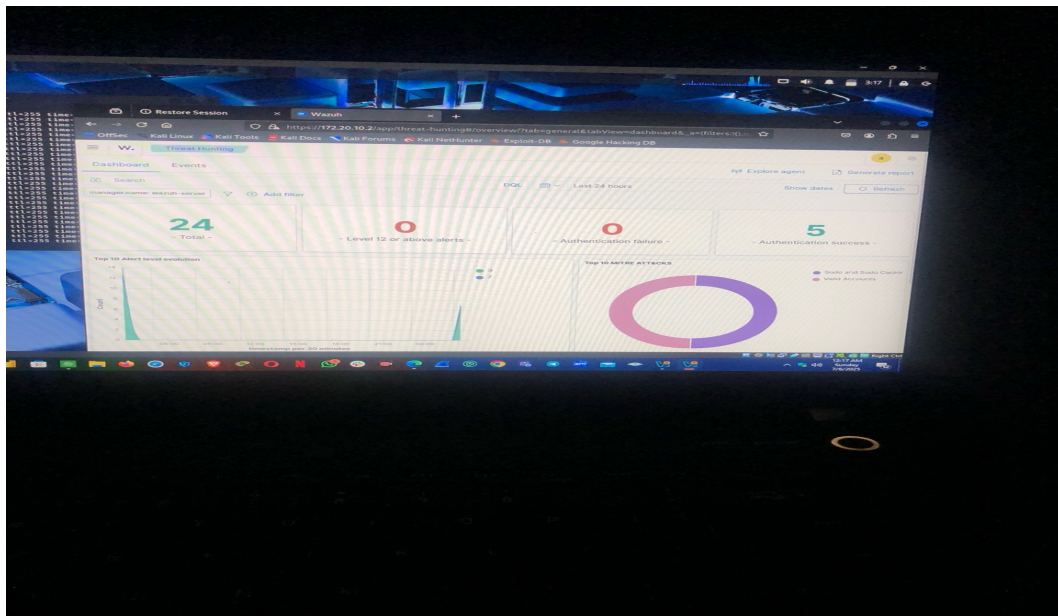


Figure 3: Detailed view of authentication and session activity logs.

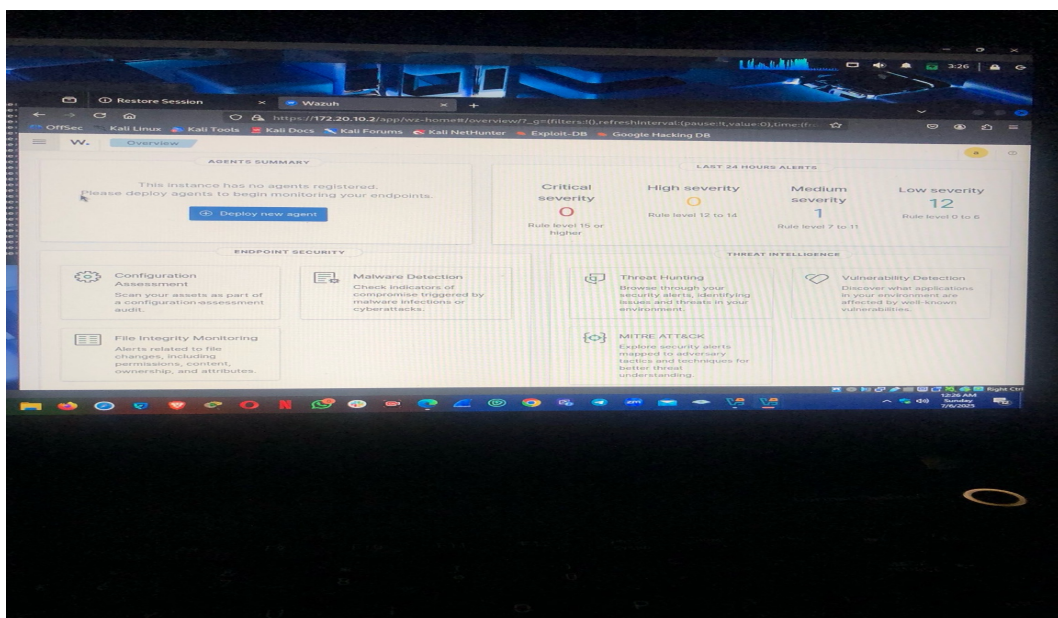


Figure 4: Privilege escalation and sudo-related alert visualization.

4. Conclusion

This project demonstrates effective use of Wazuh SIEM for host-based monitoring within a Linux environment. While no high-severity security incidents were identified, the analysis confirms that Wazuh successfully captures authentication activity, monitors privilege escalation, and provides meaningful visibility for SOC analysts. The project highlights the importance of alert context, severity assessment, and continuous monitoring in cybersecurity operations.

5. Analyst Takeaways

- SIEM alerts require contextual interpretation to avoid false positives
- Authentication and sudo logs are critical indicators of system misuse
- Dashboards enhance visibility and speed of alert triage
- Host-based monitoring is a foundational SOC capability